1

Suppose $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$. Find the approximate mean and variance of

(a) e^X
$E(e^X) \approx e^\mu$, $\text{Var}(e^x) \approx \sigma^2 e^{2\mu}$

(b) $1/X$ (assuming $\mu \neq 0$)
$E(1/X) \approx 1/\mu$, $\text{Var}(1/X) \approx \sigma^2/\mu^4$

(c) $\ln(X)$ (assuming $X > 0$).
$E(\log(X)) \approx \log(\mu)$, $\text{Var}(1/X) \approx \sigma^2/\mu^2$

2.

$$\begin{aligned}
\lim_{n\to\infty} \mathbb{E}[|X + Y_n - X|^2] &= \lim_{n\to\infty} \mathbb{E}[|Y_n|^2] \\
&= \lim_{n\to\infty} \mathbb{E}[Y_n^2] \\
&= \lim_{n\to\infty} \frac{2}{n^2} = 0.
\end{aligned}$$

3. This converges in distribution, in probability, and in first mean. Not almost surely (partial converse of Borel-Cantelli).

4.

(5.18) (c.) $f_x(x) = \underbrace{\frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p}{2}\right)\sqrt{p}\sqrt{\pi}}}_{A_p} \underbrace{\frac{1}{\left(1+\frac{x^2}{p}\right)^{(p+1)/2}}}_{K_p}$

$A_p = \downarrow$ consider normalizing constant

using Stirling's Formula $\left(n! = \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}\right)$

$$A_p = \frac{\sqrt{2\pi}\left(\frac{p-1}{2}\right)^{\frac{p-1}{2}+\frac{1}{2}} e^{-\frac{p-1}{2}}}{\sqrt{2\pi}\left(\frac{p-2}{2}\right)^{\frac{p-2}{2}+\frac{1}{2}} e^{-\frac{p-2}{2}}} \frac{1}{\sqrt{p\pi}}$$

$$= \frac{\left(\frac{p-1}{2}\right)^{p/2} e^{-p/2} e^{1/2}}{\left(\frac{p-2}{2}\right)^{p/2}\left(\frac{p-2}{2}\right)^{-1/2} e^{-p/2} e^{1} p^{1/2}\sqrt{\pi}}$$

$$= \frac{e^{-1/2}}{\sqrt{\pi}} \frac{\left(\frac{p-1}{p-2}\right)^{p/2}}{\left(\frac{p}{2}-1\right)^{-1/2}\left(\frac{1}{p}\right)^{-1/2}} = \frac{e^{-1/2}}{\sqrt{\pi}} \frac{\left(\frac{p-1}{p-2}\right)^{p/2}}{\left(\frac{1}{2}-\frac{1}{p}\right)^{-1/2}}$$

$\lim_{p\to\infty} A_p = \frac{e^{-1/2} e^{1/2}}{\sqrt{\pi}\sqrt{2}} = \frac{1}{\sqrt{2\pi}}$ since $\left(\frac{p-1}{p-2}\right)^{p/2} \longrightarrow e^{1/2}$ as $p\to\infty$ and
$\left(\frac{1}{2}-\frac{1}{p}\right)^{-1/2} \longrightarrow \left(\frac{1}{2}\right)^{-1/2} = 2^{1/2} = \sqrt{2}$ as $p\to\infty$

Now consider the kernel $k_p = \frac{1}{\left(1+\frac{x^2}{p}\right)^{(p+1)/2}} = \left(1+\frac{x^2}{p}\right)^{-p/2}\left(1+\frac{x^2}{p}\right)^{-1/2}$

$\lim_{p\to\infty} k_p = e^{-x^2/2}$ since $\left(1+\frac{x^2}{p}\right)^{-p/2} \longrightarrow e^{-x^2/2}$ as $n\to\infty$ and
$\left(1+\frac{x^2}{p}\right)^{-1/2} \longrightarrow 1$ as $n\to\infty$

so $f_x(x) \longrightarrow \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

d. let $X \sim t_p$. We showed that $X \longrightarrow N(0,1)$ as $p\to\infty$ so $X^2 \longrightarrow [N(0,1)]^2 = \chi^2_{(1)}$
Since $X^2 = F_{1,p}$ it makes sense that $F_{1,p} \xrightarrow{p\to\infty} \chi^2_1$

(e.) By (d.) $F_{1,p} \longrightarrow \chi^2_{(1)}$. $q F_{q,p} = \sum_{i=1}^{q} t_p^2 \longrightarrow \chi^2_{(q)}$

5.32

a. For any $\epsilon > 0$,

$$
\begin{aligned}
P\left(\left|\sqrt{X_n} - \sqrt{a}\right| > \epsilon\right) &= P\left(\left|\sqrt{X_n} - \sqrt{a}\right|\left|\sqrt{X_n} + \sqrt{a}\right| > \epsilon\left|\sqrt{X_n} + \sqrt{a}\right|\right) \\
&= P\left(|X_n - a| > \epsilon\left|\sqrt{X_n} + \sqrt{a}\right|\right) \\
&\leq P\left(|X_n - a| > \epsilon\sqrt{a}\right) \to 0,
\end{aligned}
$$

as $n \to \infty$, since $X_n \to$ a in probability. Thus $\sqrt{X_n} \to \sqrt{a}$ in probability.

b. For any $\epsilon > 0$,

$$
\begin{aligned}
P\left(\left|\frac{a}{X_n} - 1\right| \leq \epsilon\right) &= P\left(\frac{a}{1+\epsilon} \leq X_n \leq \frac{a}{1-\epsilon}\right) \\
&= P\left(a - \frac{a\epsilon}{1+\epsilon} \leq X_n \leq a + \frac{a\epsilon}{1-\epsilon}\right) \\
&\geq P\left(a - \frac{a\epsilon}{1+\epsilon} \leq X_n \leq a + \frac{a\epsilon}{1+\epsilon}\right) \qquad \left(a + \frac{a\epsilon}{1+\epsilon} < a + \frac{a\epsilon}{1-\epsilon}\right) \\
&= P\left(|X_n - a| \leq \frac{a\epsilon}{1+\epsilon}\right) \to 1,
\end{aligned}
$$

as $n \to \infty$, since $X_n \to$ a in probability. Thus $a/X_n \to 1$ in probability.

c. $S_n^2 \to \sigma^2$ in probability. By a), $S_n = \sqrt{S_n^2} \to \sqrt{\sigma^2} = \sigma$ in probability. By b), $\sigma/S_n \to 1$ in probability.

6.

(a.) $f_X(x) = e^{-x} \cdot F_X(x) = 1 - e^{-x}$ $\qquad F_{X_{(n)}}(x) = [1-e^{-x}]^n$

$Y_n = X_{(n)} - \log(n)$

$F_{Y_n}(y) = P(Y_n \leq y) = P(X_{(n)} - \log(n) \leq y) = P(X_{(n)} \leq$

$= F_{X_{(n)}}(y + \log(n)) = [1 - e^{-(y+\log n)}]^n$

$= [1 - e^{-y} e^{\log n^{-1}}]^n = \left[1 - \frac{e^{-y}}{n}\right]^n$

$\lim_{n\to\infty} F_{Y_n}(y) = \lim_{n\to\infty}\left[1 - \frac{e^{-y}}{n}\right]^n = e^{-e^{-y}}$

(b.) $F_X(x) = (1 - x^{-2})$ $\qquad F_{X_{(n)}}(x) = (1 - x^{-2})^n$

$Y_n = n^{-1/2} X_{(n)}$

$F_{Y_n}(y) = P(Y_n \leq y) = P(n^{-1/2} X_{(n)} \leq y) = P(X_{(n)} \leq y n^{1/2})$

$= F_{X_{(n)}}(y n^{1/2}) = [1 - (yn^{1/2})^{-2}]^n = (1 - y^{-2} n^{-1}) = \left(1 - \frac{y^{-2}}{n}\right)^n$

$\lim_{n\to\infty} F_{Y_n}(y) = \lim_{n\to\infty}\left(1 - \frac{1/y^2}{n}\right)^n = e^{-1/y^2}$

7.

Beta$(1,\beta)$ $\quad f_X(x) = \frac{\Gamma(1+\beta)}{\Gamma(1)\Gamma(\beta)} x^{1-1} (1-x)^{\beta-1} = \beta(1-x)^{\beta-1}$

$F_X(x) = \int_0^x \beta(1-t)^{\beta-1} dt = -\frac{\beta}{\beta-1+1} (1-t)^{\beta-1+1} \Big|_0^x = -(1-t)^\beta \Big|_0^x$

$= -(1-x)^\beta + (1-0)^\beta = 1-(1-x)^\beta$

$F_{X_{(n)}}(x) = [1-(1-x)^\beta]^n$

$Y_n = n^{1/\beta}(1-X_{(n)})$

$F_{Y_n}(y) = P(Y_n \le y) = P(n^{1/\beta}(1-X_{(n)}) \le y) = P(1-X_{(n)} \le yn^{-1/\beta})$

$= P(-X_{(n)} \le -1+yn^{-1/\beta}) = P(X_{(n)} > 1-yn^{-1/\beta})$

$= 1 - F_{X_{(n)}}(1-yn^{-1/\beta})$

$= 1-[1-(1-\underbrace{1+yn^{-1/\beta}}_{x})^\beta]^n = 1-[1-\frac{y^\beta}{n}]^n$

$\lim_{n\to\infty} F_{Y_n}(y) = \lim_{n\to\infty} 1-[1-\frac{y^\beta}{n}]^n = 1-e^{-y^\beta}$

5.35 a. $X_i \sim$ exponential(1). $\mu_X = 1$, Var$X = 1$. From the CLT, $\bar{X}_n$ is approximately n$(1, 1/n)$. So

$$\frac{\bar{X}_n - 1}{1/\sqrt{n}} \to Z \sim \mathrm{n}(0,1) \quad \text{and} \quad P\left(\frac{\bar{X}_n - 1}{1/\sqrt{n}} \le x\right) \to P(Z \le x).$$

b.

$$\frac{d}{dx}P(Z \le x) = \frac{d}{dx}F_Z(x) = f_Z(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}.$$

$$\begin{aligned}
&\frac{d}{dx}P\left(\frac{\bar{X}_n - 1}{1/\sqrt{n}} \le x\right) \\
&= \frac{d}{dx}\left(\sum_{i=1}^n X_i \le x\sqrt{n}+n\right) \qquad\qquad \left(W = \sum_{i=1}^n X_i \sim \mathrm{gamma}(n,1)\right) \\
&= \frac{d}{dx}F_W(x\sqrt{n}+n) = f_W(x\sqrt{n}+n)\cdot\sqrt{n} = \frac{1}{\Gamma(n)}(x\sqrt{n}+n)^{n-1}e^{-(x\sqrt{n}+n)}\sqrt{n}.
\end{aligned}$$

Therefore, $(1/\Gamma(n))(x\sqrt{n}+n)^{n-1}e^{-(x\sqrt{n}+n)}\sqrt{n} \approx \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ as $n \to \infty$. Substituting $x = 0$ yields $n! \approx n^{n+1/2}e^{-n}\sqrt{2\pi}$.

5.36 a. EY = 2 \theta, Var(Y) = 8 \theta.

$$M_Y(t) = E(e^{ty}) = E\left[E\left[e^{ty}/N=n\right]\right] = E(1-2t)^{-n}, \quad t<1/2$$

$$= \sum_{n=0}^{\infty}(1-2t)^{-n}\frac{e^{-\theta}\theta^n}{n!} = e^{-\theta}\sum_{n=0}^{\infty}\frac{[\theta/(1-2t)]^n}{n!} = e^{-\theta}e^{\theta/(1-2t)}$$

$$= e^{\frac{\theta-\theta(1-2t)}{1-2t}} = e^{2\theta t/(1-2t)} \quad t<1/2. \text{ So, let } Z = \frac{Y-2\theta}{\sqrt{8\theta}}$$

$$M_Z(t) = e^{-2\theta t/\sqrt{8\theta}} M_Y\left(\frac{t}{\sqrt{8\theta}}\right) = \exp\left[-\frac{2\theta t}{\sqrt{8\theta}} + \frac{2\theta t/\sqrt{8\theta}}{1-2t/\sqrt{8\theta}}\right]$$

$$= \exp\left\{\frac{-2\theta t}{\sqrt{8\theta}} + \frac{2\theta t}{\sqrt{8\theta}\left(\frac{\sqrt{8\theta}-2t}{\sqrt{8\theta}}\right)}\right\}$$

$$= \exp\left\{-\frac{2\theta t}{\sqrt{8\theta}} + \frac{2\theta t}{\sqrt{8\theta}-2t}\right\}$$ Taylor series (about 0) of $\frac{t}{\sqrt{8\theta}-2t}$ gives us

$$= \exp\left\{-\frac{2\theta t}{\sqrt{8\theta}} + 2\theta\left(0+\frac{t}{\sqrt{8\theta}}+\frac{t^2}{4\theta}+R\right)\right\}$$

$$= \exp\left\{-\frac{2\theta t}{\sqrt{8\theta}} + \frac{2\theta t}{\sqrt{8\theta}} + \frac{t^2}{2} + 2\theta R\right\}$$

$$= \exp\left\{\frac{t^2}{2} + 2\theta R\right\}.$$ Note

$2\theta R \to 0$ as $\theta \to \infty$ because the remainder term have θ-term to the power of $\theta^{-3/2}, \theta^{-2}, \ldots$.

Thus, $\lim_{\theta\to\infty} M_Z(t) = \exp\left\{-\frac{t^2}{2}\right\}$. $\square$

CB 5.39

(b.) In Example 5.5.8, find a subsequence of the X_is that converges almost surely, that is, that converges pointwise.

Define the subsequence $X_j(s) = s + I_{[a,b]}(s)$ such that in $I_{[a,b]}$, a is always 0. For this subsequence

$$X_j(s) \to \begin{cases} s & \text{if } s>0 \\ s+1 & \text{if } s=0. \end{cases}$$

CB 5.44 Let X_i, $i = 1, 2, \ldots$, be independent Bernoulli(p) random variables and let $Y_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

(a) Show that $\sqrt{n}(Y_n - p) \xrightarrow{d} \mathcal{N}(0, p(1-p))$.
Since $Y_n = \overline{X}_n$ and $E(\overline{X}_n) = p$, this is true by the CLT

(b) Show that for $p \neq 1/2$,

$$\sqrt{n}[Y_n(1 - Y_n) - p(1-p)] \xrightarrow{d} \mathcal{N}[0, (1-2p)^2 p(1-p)].$$

$g(p) = p(1-p) = p - p^2$ and $g'(p) = 1 - 2p$. If $p \neq 1/2$ then $g'(p) \neq 0$ so by the Delta method

$$\sqrt{n}(Y_n(1 - Y_n) - p(1-p)) \xrightarrow{d} \mathcal{N}(0, p(1-p))$$

(c) Show that for $p = 1/2$, $n[Y_n(1-Y_n) - \frac{1}{4}] \xrightarrow{d} -\frac{1}{4}\chi_1^2$. (If this appears strange, note that $Y_n(1-Y_n) \leq 1/4$, so the left-hand side is always negative. An equivalent form is $2n[\frac{1}{4} - Y_n(1-Y_n)] \xrightarrow{d} \chi_1^2$.)
Since $g'(p) = 0$ when $p = 1/2$, use the second order delta method and $g''(p) = -2 \neq 0$ so we have

$$n[Y_n(1 - Y_n) - p(1-p)] \xrightarrow{d} p(1-p)(-2/2)\chi^2_{(1)}$$

which for $p = 1/2$ works out to be

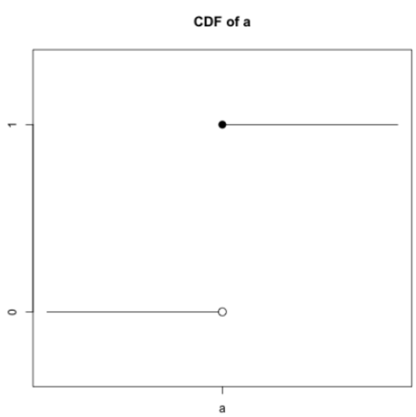
$$n[Y_n(1 - Y_n) - 1/4] \xrightarrow{d} -1/4\chi^2_{(1)}.$$

12.

Let a be a real-valued constant and $X_1, X_2, \ldots$ a sequence of random variables.

(a) Sketch the CDF of a. What does $X_n \xrightarrow{d} a$ mean?
$X_n \xrightarrow{d} a$ means that

$$F_n(x) \to F(a) = P(a \leq x) = \begin{cases} 0 & x < a \\ 1 & x > a \end{cases}$$



(b) Show that if $X_n \xrightarrow{d} a$ then $X_n \xrightarrow{p} a$. (This is "partial converse" to our implication chart–see Theorem 5.5.13.)
For $\epsilon > 0$

$$\begin{aligned}
P(|X_n - a| \geq \epsilon) &= 1 - P(|X_n - a| < \epsilon) && (1)\\
&= 1 - P(a - \epsilon < X_n < a + \epsilon) && (2)\\
&\leq 1 - P(a - \epsilon < X_n \leq a + \epsilon) && (3)\\
&= 1 - [F_{X_n}(a + \epsilon) - F_{X_n}(a - \epsilon)] && (4)\\
&\to 1 - (1 - 0) && (5)\\
&= 0 && (6)
\end{aligned}$$

The inequality in (3) is from the fact that $P(X_n < a + \epsilon) \leq P(X_n \leq a + \epsilon)$ and (5) follows since $X_n \xrightarrow{d} a$, then $F_{X_n}(a + \epsilon) \to 1$ and $F_{X_n}(a - \epsilon) \to 0$.
Hence since $P(|X_n - a| \geq \epsilon) \to 0$, $X_n \xrightarrow{p} a$.

13.

a) $P(\overline{X} > 5.33\ldots)$

$= P\left(\frac{\overline{X}-5}{2/\sqrt{40}} > \frac{5.333-5}{2/\sqrt{40}} \right)$

$\approx P\left(Z > \frac{5.33..-5}{2/\sqrt{40}} \right) = 0.1459$

b) $P(\sum_{i=1}^{n} X_i < 180)$, $\sum_{i=1}^{n} X_i \overset{\cdot}{\sim} N[200, 4.40]$

≈ 0.569

14.

$P(\sum X_i > 2000) = 1 - P(\sum X_i < 2000)$

$\approx 1 - P\left(\frac{\sum X_i - 160 \cdot 12}{30\sqrt{n}} < \frac{2000 - 12 \cdot 160}{30\sqrt{12}} \right)$

$= 0.2207$

15.

Use continuity correction

a. i) $\approx P(\sum X_i > 70.5) \overset{\text{Normal}}{\approx} 0.5874$

ii) $P(\sum X_i < 69.5) \approx 0.3563$)

iii) $P(\sum X_i \in (69.5, 70.5)) \simeq 0.0562$

b) $g(p) = \frac{p}{1-p}$, $g'(p) = \frac{1}{(1-p)^2}$. So,

$\frac{\hat{p}}{1-\hat{p}} \overset{\cdot}{\sim} N\left[\frac{p}{1-p} , \frac{p(1-p)}{n(1-p)^4} \right) = N\left[\frac{p}{1-p}, \frac{p}{n(1-p)^3} \right].$

16.

$$\sqrt{n}(\bar{X} - 50) \xrightarrow{d} N(0, 500) \text{ by CLT}$$

→ delta method gives

$r = \exp(\bar{x})$ has the following:

$$\sqrt{n}\left(\exp(\bar{x}) - \exp(50)\right) \xrightarrow{d} N\left[0, e^{100} \cdot 500\right]$$

Since $g'(\mu) = e^{\mu} = e^{50} = e^{\alpha\beta}$

Show that each of the following families is an exponential family.

(a) Normal family with parameter μ known.

Solution:

$$f(x|\mu) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-1}{2\sigma^2}(x-\mu)^2\right),$$
$$h(x) = 1,\ \ c(\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}I_{(0,\infty)}(\sigma^2),\ \ w_1(\sigma^2) = \frac{-1}{2\sigma^2},\ \ t_1(x) = (x-\mu)^2$$

(b) Normal family with parameter σ known.

Solution:

$$f(x|\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{x^2}{2\sigma^2}\right)\exp\left(-\frac{\mu^2}{2\sigma^2}\right)\exp\left(\mu\frac{x}{\sigma^2}\right),$$
$$h(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right),\ \ c(\mu) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{\mu^2}{2\sigma^2}\right),\ \ w_1(\mu) = \mu,\ \ t_1(x) = \exp\left(\frac{x}{\sigma^2}\right)$$

(c) Gamma family with parameter α known.

Solution:

$$f(x|\alpha) = \frac{1}{\Gamma(\alpha)\beta^\alpha}x^{\alpha-1}e^{\frac{-x}{\beta}},$$
$$h(x) = \frac{x^{\alpha-1}}{\Gamma(\alpha)}, x>0,\ \ c(\beta) = \frac{1}{\beta^\alpha},\ \ w_1(\beta) = \frac{1}{\beta},\ \ t_1(x) = -x$$

(d) Gamma family with parameter β known.

Solution:

$$f(x|\beta) = e^{-x/\beta}\frac{1}{\Gamma(\alpha)\beta^\alpha}\exp((\alpha-1)\log x),$$
$$h(x) = e^{-x/\beta}, x>0,\ \ c(\alpha) = \frac{1}{\Gamma(\alpha)\beta^\alpha},\ \ w_1(\alpha) = \alpha-1,\ \ t_1(x) = \log x$$

(e) Gamma family with both α and β unknown.

Solution:

$$f(x|\alpha,\beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha}\exp((\alpha-1)\log x - \frac{x}{\beta}),$$
$$h(x) = I_{(x>0)}(x),\ \ c(\alpha,\beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha},\ \ w_1(\alpha) = \alpha-1,\ \ t_1(x) = \log x,\ \ w_2(\alpha,\beta) = -1/\beta,\ \ t_2(x) = x$$

(f) Beta family with parameter α known.

Solution:

$$h(x) = x^{\alpha-1}I_{[0,1]}(x),\ \ c(\beta) = \frac{1}{B(\alpha,\beta)},\ \ w_1(\beta) = \beta-1,\ \ t_1(x) = \log(1-x).$$

(g) Beta family with parameter β known.

Solution:

$$h(x) = (1-x)^{\beta-1}I_{[0,1]}(x),\ \ c(\alpha) = \frac{1}{B(\alpha,\beta)},\ \ w_1(\alpha) = \alpha-1,\ \ t_1(x) = \log x.$$

(h) Poisson family.

Solution:

$$h(x) = \frac{1}{x!}I_{(0,1,2\ldots)}(x),\ \ c(\theta) = e^{-\theta},\ \ w_1(\theta) = \log\theta,\ \ t_1(x) = x.$$

(a) Negative Binomial family with r known, $0 < p < 1$.

Solution:

$$h(x) = \binom{x-1}{r-1} I_{(r,r+1,\dots)}(x), \;\; c(p) = \left(\frac{p}{1-p}\right)^r \quad w_1(p) = \log(1-p), \quad t_1(x) = x$$

(b) Pareto with α known.

Solution:

$$h(x) = I_{(\alpha,\infty)}(x), \;\; c(\theta) = \beta\alpha^\beta \quad w_1(\theta) = -(\beta+1), \quad t_1(x) = \log(x)$$

(c) Beta family with both α and β unknown.

Solution:

$$h(x) = I_{[0,1]}(x), \;\; c(\alpha,\beta) = \frac{1}{B(\alpha,\beta)}, \;\; w_1(\alpha) = \alpha - 1, \;\; t_1(x) = \log x,$$
$$w_2(\beta) = \beta - 1, \;\; t_2(x) = \log(1-x).$$

(a.) gamma(α, β)

$$f(x) = \left(\frac{1}{\Gamma(\alpha)\beta^{\alpha}}\right)\left(e^{(\alpha-1)\log x + -\frac{1}{\beta}x}\right)$$

$\eta_1 = (\alpha-1)$ $\quad \eta_2 = -\frac{1}{\beta}$

$\alpha = \eta_1 + 1$ $\quad \beta = -\frac{1}{\eta_2}$

$$C(\eta) = \frac{1}{\Gamma(\eta_1+1)(-\frac{1}{\eta_2})^{(\eta_1+1)}}$$

$$= \left[\Gamma(\eta_1+1)\right]^{-1} \left(\frac{1}{-\eta_2}\right)^{-(\eta_1+1)} = \left[\Gamma(\eta_1+1)\right]^{-1} (-\eta_2)^{(\eta_1+1)}$$

$$\log c(\eta) = -1 \log \Gamma(\eta_1+1) + (\eta_1+1)\log(-\eta_2)$$

$$\frac{\partial \log c(\eta)}{\partial \eta_2} = \frac{\eta_1+1}{(-\eta_2)} \cdot (-1)$$

since $t_2 = x$ take derivative with respect to η_2

$$\frac{-\partial \log c(\eta)}{\partial \eta_2} = -\left(\frac{\eta_1+1}{\eta_2}\right)$$

$$\frac{-\partial^2 \log c(\eta)}{\partial \eta_2^2} = +\frac{\eta_1+1}{\eta_2^2}$$

$$E(x) = \frac{\eta_1+1}{\eta_2} = \left(\frac{(\alpha-1)+1}{-\frac{1}{\beta}}\right) = \alpha\beta$$

$$\text{var}(x) = \frac{\eta_1+1}{\eta_2^2} = \frac{\alpha-1+1}{(\frac{1}{-\beta})^2} = \alpha\beta^2$$

$$f(x) = \left(\frac{1}{x!}\right) e^{-\lambda} e^{x \log \lambda}$$

$\eta = \log \lambda$

$\lambda = e^{\eta}$

$$C(\eta) = e^{-e^{\eta}}, \quad \log C(\eta) = -e^{\eta}$$

$$-\frac{\partial \log c(\eta)}{\partial \eta} = +e^{\eta}$$

$$-\frac{\partial^2 \log c(\eta)}{\partial \eta^2} = e^{\eta}$$

$$E(x) = e^{\eta} = e^{\log \lambda} = \lambda$$

$$\text{Var}(x) = e^{\eta} = e^{\log \lambda} = \lambda$$

3.33 a. (i) $h(x) = e^x I_{\{-\infty<x<\infty\}}(x)$, $c(\theta) = \frac{1}{\sqrt{2\pi\theta}} \exp(\frac{-\theta}{2})\theta > 0$, $w_1(\theta) = \frac{1}{2\theta}$, $t_1(x) = -x^2$.

(ii) The nonnegative real line.

b. (i) $h(x) = I_{\{-\infty<x<\infty\}}(x)$, $c(\theta) = \frac{1}{\sqrt{2\pi a\theta^2}} \exp(\frac{-1}{2a}) - \infty < \theta < \infty, a > 0$,
$w_1(\theta) = \frac{1}{2a\theta^2}$, $w_2(\theta) = \frac{1}{a\theta}$, $t_1(x) = -x^2$, $t_2(x) = x$.

(ii) A parabola.

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c. (i) $h(x) = \frac{1}{x} I_{\{0<x<\infty\}}(x)$, $c(\alpha) = \frac{\alpha^\alpha}{\Gamma(\alpha)}\alpha > 0$, $w_1(\alpha) = \alpha$, $w_2(\alpha) = \alpha$,
$t_1(x) = \log(x)$, $t_2(x) = -x$.

(ii) A line.

d. (i) $h(x) = C\exp(x^4) I_{\{-\infty<x<\infty\}}(x)$, $c(\theta) = \exp(\theta^4) - \infty < \theta < \infty$, $w_1(\theta) = \theta$,
$w_2(\theta) = \theta^2$, $w_3(\theta) = \theta^3$, $t_1(x) = -4x^3$, $t_2(x) = 6x^2$, $t_3(x) = -4x$.

(ii) The curve is a spiral in 3-space.